2DLE

2D Level Editor Engine

v2.0 Complete Developer & Scripting Guide

J2ME / MIDP 2.0 Mobile RPG Engine

87 Tile Types • 35 Script Commands • .2dip Distribution

1. Engine Overview

2DLE is a complete 2D RPG engine for J2ME (MIDP 2.0) mobile devices. It includes a full tile-based level editor, a scripting language (2DLS), a battle system, NPC management, weather effects, day/night cycles, map transitions, and a distribution format (.2dip) that packages entire games for public release.

1.1 Architecture — All 39 Source Files

Class	Lines	Responsibility
`EditorCanvas`	4500	Main editor: tile paint, NPC placement, scripting, playtest, export
`LevelEditorMIDlet`	300	App lifecycle, screen switching, game launch
`PlayEngine`	987	Player movement, sliding, particles, dialogue, damage
`BattleSystem`	1053	Turn-based / ATB combat, skills, status effects
`EventSystem`	~600	Script bytecode interpreter (runs compiled .2dls)
`ScriptLang`	961	Text → bytecode compiler, decompiler, templates
`ScriptManager`	458	Script registry: maps tile/NPC IDs to bytecodes
`ScriptEditor`	542	In-editor script writing and compile UI
`MapLinkSystem`	311	Door/stairs/warp transitions between levels
`LinkEditor`	268	UI for configuring map link targets and spawn points
`BuiltinNPCs`	470	15 pre-made characters (10 NPC + 5 boss) with scripts
`GameCompiler`	370	Packages project → .2dip binary (levels+assets+scripts)
`GamePlayer`	830	Standalone runtime: plays .2dip games, title screen, save
`GamePlayerMIDlet`	221	Public distribution MIDlet — picks .2dip from storage
`TileData`	546	87 tile types with collision, animation, light, encounter data
`NPCManager`	875	NPC positions, types, dialogue per level
`EnemyData`	~400	10 enemy types with AI, elements, stats
`PlayerData`	641	Player stats, level, HP/MP, class, serialization
`InventorySystem`	~500	Items, equipment, gold, shops
`SkillSystem`	~450	20 skills: physical/magic/heal/buff/debuff
`QuestSystem`	~400	80 quests with 5 objective types
`WeatherSystem`	~350	8 weather types with particle effects
`DayNightSystem`	~300	24-hour cycle, seasons, moon phases
`FactionSystem`	~350	10 factions with reputation ripple effects
`StealthSystem`	~250	Stealth level, tile modifiers, detection
`DungeonGenerator`	~400	Procedural BSP dungeon rooms and corridors
`SoundManager`	~300	32 SFX + 21 music, 4-channel pool, positional audio
`FileManager`	1209	Storage detect, backup/restore, atomic write
`ProjectManager`	~600	Project CRUD, splash screen, install chooser

`WorldMap`	~300	Overworld map with fog of war
`AISystem`	~350	Enemy AI: patrol, chase, flee, guard patterns
`CutsceneSystem`	~400	Scripted cutscene playback with camera moves
`ShopSystem`	~300	Buy/sell UI with haggling
`CraftingSystem`	~300	Recipe-based item crafting
`DialogueEditor`	~350	In-editor NPC dialogue tree editor
`GameConfig`	387	All engine constants and difficulty presets

2. Level Editor Controls

2.1 Key Bindings

`2/4/6/8`	Move cursor on the map (or D-pad)
`5`	Apply draw tool / interact / place NPC (title screen: open builtin picker)
`0`	Open Script Editor for tile or NPC under cursor
`1 / 3`	Previous / next tile (or NPC type in NPC mode)
`#`	On door/warp tile: open Link Editor. Otherwise: cycle layer → NPC mode
`*`	Toggle grid / cancel shape drag / exit playtest
`7 / 9`	Cycle draw tool backward / forward (14 tools total)
`Softkey L/R`	Zoom in (x1→x2→x3) and zoom out
`Menu`	Open menu (Save, Load, New Map, Export .2dip, Play .2dip, Exit…)

2.2 Layers

- Layer 0 — BG: walkable ground tiles. Step scripts and encounter triggers fire here.
- Layer 1 — Wall: collision tiles (brick, tree, castle wall). Blocks player movement.
- Layer 2 — Obj: foreground objects (chests, doors, signs, transport).
- NPC Mode (# until "NPC Mode"): press 5 = Builtin NPC Picker; press 0 = Script Editor.

2.3 New Map — Safe Create

Menu → New Map auto-saves the current level first, then asks for a new level name. Never overwrites anything silently. Give every level a unique name (e.g. "village", "dungeon1", "inn") — map links reference levels by name.

3. Tile Reference (87 Types)

All 87 tile IDs listed below. Select in editor with 1/3 keys. Animated tiles cycle at ~5fps.

3.1 Ground & Nature (IDs 0–39)

0 Void	Empty — transparent on layers 1-2
1-3 Grass	Walkable ground, dust particles on walk
4 Dirt	Walkable, speckled pattern
5 Stone	Solid walkable stone
6 Rock	Solid obstacle, can be pushed
7 Water	2-frame wave animation. Player swims (+50% move delay)
8 Brick	Solid wall. Use on Layer 1
9 Wood/Floor	Walkable wooden floor with grain
10 Chest	Interact trigger. Attach script to give items
11 Lava	Damage on each step. COL_DAMAGE flag
13 Dirt Path	Slightly darker walkable path
14 Floor Tile	Checkerboard stone floor tile
16 Tree/Bush	Solid obstacle with canopy + trunk
17 Flower	Walkable, pink/red flower dots on grass
21 Door	Interact or step trigger. Attach Link for map transition
22/23 Stairs	Step trigger. Link up or down to another floor
25 Sign	Interact trigger. Attach script to show text
26 Fence	Solid obstacle, vertical bar pattern
31 Ice	COL_SLIDE — player slides in movement direction
32 Swamp	2-frame. COL_SLOW + poison chance
33 Quicksand	2-frame. Sink trap
34 Mud	COL_SLOW, dirty footprints
35 Tall Grass	+30% random encounter rate
36 Telepad	2-frame. Step-triggered instant warp. Link to any level
37 Cactus	Solid, damage on contact
38 Coral	Decorative, underwater
39 Cloud	Walkable sky tile

3.2 Interactive & Hazard (IDs 40–65)

40 Spike Trap	COL_DAMAGE, hidden until stepped on
41 Barrel	Pushable obstacle
42 Bookshelf	Interact trigger, can hold clue scripts
43 Fireplace	2-frame fire animation, emits warm light
44 Altar	Save point + full heal on interact
45 Warp Stone	2-frame. Fast-travel anchor, manual trigger (press 5)
46 Cracked Wall	Destructible — blow up with bomb item

47 Deep Water	2-frame. Requires boat item to cross
48-51 Conveyors	Moves player automatically R/L/U/D
52 Bomb Wall	Destructible tile — bomb item required
53 Torch	2-frame. Wall sconce, emits light 120
54 Crystal Ball	2-frame animated orb, interact trigger
55 Trap Floor	Hidden until stepped on, drops player
56 Vine	Climbable + slow, vertical surface
57 Glowing Orb	2-frame. Permanent light source (200 intensity)
58 Mirror	Reflect projectiles, interact trigger
59 Seabed	Underwater floor tile
60 Night Sky	2-frame animated starfield tile
61 Volcano	2-frame lava eruption tile
62 Crystal	2-frame sparkling mineral tile
63 Cave	Dark cave entrance tile with purple tones
64 House Door	Auto-step trigger. Attach link to interior level
65 House Exit	Auto-step trigger. Attach link back to outdoor level

3.3 NEW — Buildings (IDs 66–73)

66 Shop	2-frame animated sign. COL_INTERACT. Attach shop script
67 Inn	Lit windows. COL_INTERACT. Attach inn rest script
68 Castle Wall	Solid. Battlement pattern. Use on layer 1
69 Castle Gate	Portcullis bars. COL_INTERACT. Attach link for entry
70 Tower	Solid round tower top. Emits dim light
71 Ruin	Crumbled stone wall, decorative
72 Temple	2-frame glow. COL_INTERACT. Attach healing/quest script
73 House Wall	Side wall with window. Solid, use layer 1

3.4 NEW — Transport (IDs 74–78, 4-frame animated)

74 Boat	Sail sways left/right. COL_INTERACT. Step on to board
75 Cart	Wheel spins 4 positions. COL_INTERACT + COL_PUSHABLE
76 Horse	Full trot cycle. COL_INTERACT. Attach mount script
77 Airship	Propeller spins. COL_INTERACT. Attach fly script
78 Raft	Bobs on water. COL_INTERACT. Attach sail script

3.5 NEW — Animals (IDs 79–86, 4-frame animated)

79 Bird	Wing flap cycle. COL_NONE — decorative
80 Fish	Tail wag swim. COL_NONE — use on water tiles
81 Cat	Tail twitch idle. COL_INTERACT — meow script
82 Dog	Tail wag walk. COL_INTERACT — bark/follow script
83 Rabbit	Hop bounce. COL_NONE — decorative wildlife
84 Deer	Walking gait. COL_NONE — +20% encounters
85 Wolf	Prowl cycle. COL_DAMAGE — +30% encounters
86 Bear	Stomp cycle. COL_DAMAGE — +40% encounters

4. Map Links — Doors, Stairs & Warps

Map links connect levels together. When the player steps on or interacts with a linked tile in playtest/game mode, the engine saves the current position and loads the target level, placing the player at the configured spawn point.

4.1 Creating a Link

- 1. Place a door/stairs/telepad/house-door tile on the map.
- 2. Move cursor onto that tile and press #.
- 3. In the Link Editor: set target level name, spawn X/Y, facing direction, trigger mode.
- 4. Press Save Link. A cyan border + arrow appears on the tile in the editor.

4.2 Trigger Modes

Auto (step)	Player transitions as soon as they walk onto the tile (doors, telepads, edges)
Manual (5 key)	Player must press 5 while facing the tile (gates, warp stones, NPCs)

4.3 House Pattern

```
Outside level:  TILE_HOUSE_DOOR at (5,3) → target="inn"  spawnX=4 spawnY=7  Auto
Inside level:   TILE_HOUSE_EXIT at (4,7) → target="village" spawnX=5 spawnY=4 Auto
```

4.4 Dungeon Stairs Pattern

```
Floor 1:  TILE_STAIRS_DN at (10,8) → target="dungeon_b2"  spawnX=5  spawnY=2
Floor 2:  TILE_STAIRS_UP at (5,2)  → target="dungeon_b1"  spawnX=10 spawnY=7
```

4.5 Edge Connection (Seamless Maps)

Place an EdgeN/S/E/W link (no tile needed — set via # then choose Edge type) to make the player seamlessly walk from one map to the adjacent one, like connected overworld zones.

5. NPC System & Built-in Characters

5.1 Placing NPCs

- Press # until HUD shows "NPC Mode".
- Press 5 on an empty tile → Builtin NPC Picker opens (15 characters).
- Select any NPC or Boss → placed with full default script pre-attached.
- Press 0 on an NPC → Script Editor opens to customise the script.

5.2 Built-in NPC Library

ID	Name	Type	Script Summary
0	`Villager`	Friendly	Greet on first meet (switch guard), give 5 EXP
1	`Guard`	Friendly	Block passage, 2-choice dialogue, unlock on accept
2	`Merchant`	Shopkeeper	2-choice buy/leave, opens shop 0
3	`Innkeeper`	Friendly	Rest for 10 gold, full HP/MP heal
4	`Elder`	Quest	Lore text, give 2 Potions on first meet
5	`Blacksmith`	Shopkeeper	3-choice: weapons (shop 1), craft, leave
6	`Healer`	Friendly	Free heal once, then costs 20 gold
7	`Quest Giver`	Quest	Full quest: accept/decline/complete/reward
8	`Knight`	Friendly	Training battle (Skeleton lv3), gives Axe
9	`Wizard`	Quest	Teach Fire or Ice skill for 50 gold each
10	`Slime King`	BOSS	Boss fight lv+3, 200 EXP, one-shot switch
11	`Orc Warlord`	BOSS	Boss fight lv+5, 200 EXP+gold, Axe reward
12	`Lich`	BOSS	Storm weather + fadeout + boss, 350 EXP
13	`Dragon Lord`	BOSS	Final boss, sets SW_FINAL_BOSS_DONE, 500g+EXP
14	`Dark Knight`	BOSS	Story boss with dialogue branch, gives rare item

6. 2DLS Scripting Language

2DLS (2D Level Script) is a line-based language compiled to bytecode by ScriptLang.java and executed by EventSystem.java. Press 0 on any tile or NPC to open the Script Editor.

6.1 Script Attachment Points

`Tile script`	Fires when player STEPS ON that tile type (layer 0). All same-type tiles share one script.
`NPC script`	Fires when player INTERACTS with that specific NPC slot (press 5 facing the NPC).
`Map script`	Fires on map LOAD (use for cutscenes, weather setup, boss music).
`Common script`	Subroutine called by other scripts via callCommon N.

6.2 Full Command Reference

Dialogue

`text "msg"`	Show a dialogue box. Press 5 to advance.
`choice N "a" "b" "c"`	Show N options. Result in var[0]: 0=first choice.
`wait 60`	Pause script for 60 frames (≈1 second at 60fps).

Variables & Switches

`set var 5 10`	var[5] = 10
`add var 5 3`	var[5] += 3
`sub var 5 1`	var[5] -= 1
`mul var 5 2`	var[5] *= 2
`div var 5 2`	var[5] /= 2
`mod var 5 3`	var[5] %= 3
`rnd var 5 100`	var[5] = random 0..99
`set switch 3 on`	switch[3] = true
`set switch 3 off`	switch[3] = false

Conditionals & Flow

`if switch 3 on goto label`	Jump if switch[3] is true
`if switch 3 off goto label`	Jump if switch[3] is false
`if var 5 == 10 goto label`	Jump on condition. Ops: == != > < >= <=
`if hasitem 0 2 goto label`	Jump if player has ≥2 of item[0]
`if gold >= 500 goto label`	Jump if gold ≥ 500
`label my_label`	Define jump target
`goto my_label`	Unconditional jump
`loop / endloop / break`	Loop block with break support
`end`	End script execution (REQUIRED)

Items, Gold, EXP

`give item 0 3`	Give 3 of item[0]

`take item 0 1`	Remove 1 of item[0]
`give gold 500`	Add 500 gold
`take gold 200`	Remove 200 gold
`give exp 100`	Add 100 EXP (triggers level-up)
`give skillpt`	Give 1 skill point

Player

`heal`	Full HP+MP restore
`damage 25`	Deal 25 damage to player
`status 1`	1=poison 2=burn 4=freeze 8=sleep…
`learnSkill 4`	Teach skill ID 4 (Fire) to player
`changeClass 2`	Switch player class to ID 2

World & Environment

`teleport 2 10 5`	Warp to map ID 2 at tile (10,5)
`weather 4 90`	Set weather: 0=clear 1=rain 2=fog 3=snow 4=storm 5=sand
`shake 3 30`	Screen shake intensity 3 for 30 frames
`fadeout 4`	Fade to black (speed 1-10)
`fadein 4`	Fade in from black
`sound 3`	Play SFX ID 3
`music 2`	Play BGM ID 2

Combat & Game Flow

`battle 2 0 3`	Start battle: 2 enemies of types 0 and 3
`shop 1`	Open shop ID 1
`inn`	Open inn rest screen
`craft`	Open crafting station
`savepoint`	Trigger save prompt
`cutscene 2`	Play cutscene ID 2
`startQuest 5`	Activate quest ID 5
`checkQuest 5 done_label`	Jump if quest 5 complete
`updateQuest 5 1`	Add 1 progress to quest 5

6.3 Complete Script Examples

Treasure Chest (One-Shot)

```
if switch 10 on goto empty
sound 10
text "You found the Ancient Sword!"
give item 10 1
give exp 30
set switch 10 on
goto done
label empty
text "(The chest is empty.)"
label done
end
```

Shop NPC with Gold Check

```
text "Welcome! Finest goods in the land."
choice 2 "Browse" "Leave"
if var 0 == 1 goto leave
shop 0
text "Thank you!"
goto done
label leave
text "Safe travels!"
label done
end
```

Boss Battle with Weather + Music

```
if switch 50 on goto dead
text "The Dragon awakens!"
shake 5 40
weather 4 100
fadeout 3
music 4
fadein 3
battle 1 9
set switch 50 on
set switch 99 on  // SW_FINAL_BOSS_DONE
weather 0 0
music 5
text "The dragon falls..."
give exp 500
give gold 500
goto done
label dead
text "The air is silent here."
label done
end
```

7. .2DIP Game Distribution Format

The .2dip format (2D Interactive Package) bundles your entire game — all levels, assets, scripts, NPC data, and metadata — into a single binary file for distribution to players.

7.1 What Gets Packed

LEVEL section	One per saved level. Contains map tile data + NPC positions.
LINKS section	One per level. Contains all map transition links for that level.
SCRIPTS section	One file: all compiled script bytecodes (scripts.sreg).
NPC section	All NPC dialogue text exported from NPCManager.
ASSET section	Every file in the project assets/ folder (PNG sprites, icons, sounds).
ICON section	icon.png or icon.raw — shown on title screen.
META section	Key=value metadata: title, author, version, levelCount, assetCount.

7.2 How to Compile and Publish

- Step 1: Design all levels, scripts, and NPC dialogues in the editor.
- Step 2: Place any custom sprites (hero.png, npc0.png, etc.) in the project assets/ folder.
- Step 3: Save all levels (Menu → Save Map for each level).
- Step 4: Menu → Export .2dip — fill in title, author, version, and start level name.
- Step 5: Press Build .2dip. The status bar shows: "OK: N levels M assets X KB".
- Step 6: The .2dip file is saved to 2DLE/<project>/publish/<title>.2dip.
- Step 7: Copy the GamePlayerMIDlet.jar and the .2dip file to the target device.
- Step 8: Launch GamePlayerMIDlet — it auto-scans storage and lists all .2dip games.

TIP: For a fully self-contained game JAR, rename your .2dip to game.2dip and place it inside the GamePlayerMIDlet JAR as a resource. The player MIDlet will auto-load it on startup without needing separate files.

7.3 Running .2dip from the Editor

Menu → Play .2dip Game scans the project publish/ folder and all device storage for .2dip files. Select one to launch it in GamePlayer full-screen. Controls:

2/4/6/8	Move player
5	Interact with NPC, door, or tile
*** (once)**	Pause — shows "* again = exit" hint
*** (twice)**	Exit game — returns to the editor with "Returned from game." status
0	Quick-save game state to RecordStore

7.4 Asset Loading at Runtime

GamePlayer loads all ASSET sections from the .2dip into an in-memory image cache on startup. Custom sprites are retrieved by filename:

hero.png	Player character sprite (used instead of default white square)
npc0.png	Sprite for NPC type 0 (Friendly)
npc1.png	Sprite for NPC type 1 (Shopkeeper)
icon.png	Game icon shown on title screen (any size, centered above title)

`tileset.png`	Custom tile override sheet (replaces generated pixel art)

⚠ **NOTE:** Asset filenames are case-sensitive on device storage. Use lowercase names in your scripts and assets/ folder.

8. Pixel Art Tile Rendering

Every tile is rendered as an 8×8 pixel-art sprite using a palette system, then nearest-neighbour scaled to the editor tile size (supports x1, x2, x3 zoom). All 87 tiles have unique hand-crafted sprites.

8.1 Sprite Palette System

Each sprite uses single-character colour codes mapped to a 11-colour palette generated from the tile's base colour in TileData.TILE_COLORS:

`' ' (space)`	Base colour from TileData.TILE_COLORS
`'L'`	Lighter (+40 brightness)
`'l'`	Lightest (+70 brightness)
`'D'`	Darker (-40 brightness)
`'d'`	Darkest (-70 brightness)
`'0'`	Fixed black (#111111)
`'W'`	Fixed white (#EEEEEE)
`'1' '2' '3'`	Accent colours (per-tile custom: e.g. gold, fire-orange, moss-green)

8.2 Animation

Animated tiles use a String[] array with 2 or 4 frames. The editor cycles them using animFrame % frameCount at ~5fps. Transport and animal tiles use 4 frames for smooth movement cycles; water/lava/fire use 2 frames.

9. Project Structure and Saving

9.1 File Layout on Device Storage

```
2DLE/
  <projectname>/
    level01.l2de          map tile + NPC data
    level02.l2de
    links_level01.mls     map transition links for level01
    links_level02.mls
    scripts.sreg          compiled script registry
    npcs.txt              NPC dialogue export
    assets/               sprites, icons, sounds
      hero.png
      icon.png
    publish/              compiled games
      MyGame.2dip         complete distributable package
    backups/              auto-backup archives (.2dlebak)
```

9.2 Auto-Save

The editor auto-saves before New Map, before Export .2dip, and before launching Play .2dip. Manual save: Menu → Save Map.

10. Common Pitfalls & Solutions

`Script has no end`	Always end every script with the end command.
`Door not triggering`	The tile must have a map link. Press # on it in editor.
`Link to wrong level`	Level name in Link Editor must exactly match the filename (without .l2de).
`Assets missing in .2dip`	Put ALL sprites/images in the project assets/ folder before exporting.
`icon.png not showing`	Name it exactly icon.png (lowercase) and place it in assets/
`Script tile fires every step`	Add: if switch N on goto done / ... / set switch N on / label done / end
`NPC script not firing`	Must be in NPC mode (# until "NPC Mode") and cursor on the NPC.
`New Map wiped my level`	It auto-saves first. Your level is safe — check the level list.
`binary literals 0b...`	Not valid in J2ME 1.3. Use decimal or hex (0xFF) instead.
`String.replace(String,String)`	Not in CLDC 1.0. Use substring/endsWith workarounds.
`Game exits immediately`	Start level name in Export form must match an existing saved level.

Appendix A: Sound Effect IDs

`SND_CURSOR=0`	UI cursor move
`SND_CONFIRM=1`	Menu confirm
`SND_CANCEL=2`	Menu cancel
`SND_ATTACK=3`	Physical swing
`SND_HIT=4`	Take damage
`SND_HEAL=5`	HP restore
`SND_MAGIC=6`	Spell cast
`SND_VICTORY=7`	Win jingle
`SND_DEFEAT=8`	Game over sting
`SND_LEVELUP=9`	Level up
`SND_CHEST=10`	Chest open
`SND_DOOR=11`	Door open
`SND_THUNDER=21`	Thunder crack

SND_LIMIT=26	Limit Break

Appendix B: Music Track IDs

MUS_TITLE=0	Title screen
MUS_FIELD=1	Overworld
MUS_BATTLE=2	Combat
MUS_DUNGEON=3	Dungeon
MUS_BOSS=4	Boss battle
MUS_VICTORY=5	Victory
MUS_GAMEOVER=6	Game over
MUS_TOWN=7	Town/Village
MUS_CAVE=9	Cave
MUS_FOREST=10	Forest
MUS_TENSION=16	Dramatic
MUS_FANFARE=17	Item get

Appendix C: Item IDs

0 Potion	Heal 50 HP
1 Hi-Potion	Heal 150 HP
2 Ether	Restore 30 MP
3 Antidote	Cure poison
4 Phoenix	Revive at 50% HP
5 Tent	Full restore
10 Sword	+5 ATK
11 Axe	+8 ATK
12 Staff	+3 ATK magic
20 Leather	+3 DEF
21 Chainmail	+6 DEF
22 Plate	+10 DEF